

Zihao Jing

Western University, Canada

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I am a Master's student in Computer Science at Western University, Canada. My work focuses on LLM post-training, multimodal alignment, and model adaptation. I have first-author publications at NeurIPS, ICLR, and ICML during my Master's, and prior industry experience in text embedding and generative models at SenseTime.

EDUCATION

Western University, M.Sc. in Computer Science, Ontario, Canada	Sep. 2024–Jun. 2026
Beihang University, B.Eng. in Software Engineering, Beijing, China	Sep. 2019–Jun. 2024

SELECTED PUBLICATIONS

Top-Tier Conferences

- [1] **Zihao Jing**, Qiuhaio Zeng, Ruiyi Fang, Yan Yi Li, Yan Sun, Boyu Wang, Pingzhao Hu. *Scaling-Aware Adapter for Structure-Grounded LLM Reasoning*. **ICML 2026** [Code] [Paper]
- Designed an entropy-guided smart adapter that focuses on the instruction-related parts of 3D entities, solving the longstanding gap of LLMs treating all positions as equally important. Achieves top-1 on 17 out of 18 biology QA benchmarks, making LLM-assisted structural research an operational reality.
- [2] **Zihao Jing**, Qiuhaio Zeng, Ruiyi Fang, Yan Sun, Boyu Wang, Pingzhao Hu. *Entropy-Guided Dynamic Tokens for Graph-LLM Alignment in Molecular Understanding*. In **ICLR 2026**. [Code] [Paper] [Poster] [Video][Slices]
- A dynamic connector that scales representation slots based on complexity, bridging the gap of fixed-size connectors that lose structural detail for large entities. The model ranks #1 on 20 of 21 understanding tasks and trains 3.5× faster under equivalent settings, directly advancing AI-driven drug design.
- [3] **Zihao Jing**, Yan Sun, Yan Yi Li, Sugitha Janarthanan, Alana Deng, Pingzhao Hu. *Structure-Aware Fusion with Progressive Injection for Multimodal Molecular Representation Learning*. In **NeurIPS 2025**. [Code] [Paper] [Video&Poster]
- Built a multimodal encoder that first stabilizes noisy 3D shape data into a reliable prior, solving the longstanding instability in multimodal embedding that incorporates 3D geometry. Ranks #1 on 22/29 molecular property benchmarks with up to 27% better accuracy, and was applied to a pipeline for finding Pin1 inhibitors in cancer treatment.

Selected Additional Works

- [4] Junqin Huang, Zhongjie Hu, **Zihao Jing**, Mengya Gao, Yichao Wu. *Piccolo2: General Text Embedding with Multi-Task Hybrid Loss Training*. **SenseTime Technical Report, 2024** [SenseNova] [Code] [Report]
- Built training/evaluation pipelines and contributed to iterative optimization of a general-purpose embedding model; achieved top-1 on C-MTEB in May 2024.

Additional Co-authored Publications

Co-author on 3 additional papers at **ICML 2026** and **ICLR 2026** (×2).

RESEARCH GRANTS

Digital Research Alliance of Canada, RRG Competition 2026 — Secured 5×A100-80GB GPU-years on Canada's national supercomputing cluster (~US\$80K commercial value).

RESEARCH INTERNSHIPS

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|---|----------------------------|-----------------|
| SenseTime | <i>LLM Research Intern</i> | 2023.09–2024.06 |
| <ul style="list-style-type: none">Developed Piccolo-GPT, supporting both text embedding and generation within a single LLM architecture.Trained general-purpose text embedding models (Piccolo2); top-1 on C-MTEB (May 2024).Fine-tuned a ~100B LLM for vertical livestream marketing for Sina Weibo. | | |
| Jina AI (acquired by Elastic) | <i>AI Research Intern</i> | 2023.04–2023.09 |
| <ul style="list-style-type: none">Implemented LLM-based denoising and sentiment analysis pipeline for Budweiser; reduced cost >13%.Led evaluation and deploying of super-resolution models for commercialization.Contributed to RunGPT of LlamaIndex with our super-resolution pipeline. | | |

TECHNICAL SKILLS

- Research:** LLM post-training, pre-training/fine-tuning, and multimodal alignment.
- ML Engineering:** PyTorch, Transformers, DeepSpeed, DDP/FSDP, and multi-GPU/node training.
- Systems and Programming:** Linux, Slurm, Docker, HPC workflows, Python, C/C++, Java, and SQL.

SELECTED AWARDS AND HONORS

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| • Third Prize , 13th National Undergraduate Mathematics Competition (National-level) | 2021 |
| • Third Prize , 32nd Beijing Undergraduate Mathematics Competition (Province-level) | 2021 |
| • Honorable Mention (H Prize) , Mathematical Contest In Modeling | 2020 |